

tnbc-92

three-hypothesis evaluation framework

what each hypothesis measures, how the metrics work mathematically, and what the results mean

scope	Wang et al. 2024 TNBC cohort 94 patients . 260 matched subarrays . ~278k niches
unit	niche (k=6 spatial neighbors), ~1200 cells
evaluation	subarray-level, patient-stratified 85/15 split
permutations	seed=42, 1e5 perms for H3-extended (Bareche labels), 1e3 perms for H3-proposal (per-pathway CCA)
FDR control	BH-FDR over 20-test family (H3-extended), 120-test grid (H3 5-parent), 9480-cell grid (H3 DAG)
affiliation	OHSU Knight Cancer Institute . Creason Lab
funding	ResearchHub Foundation grant

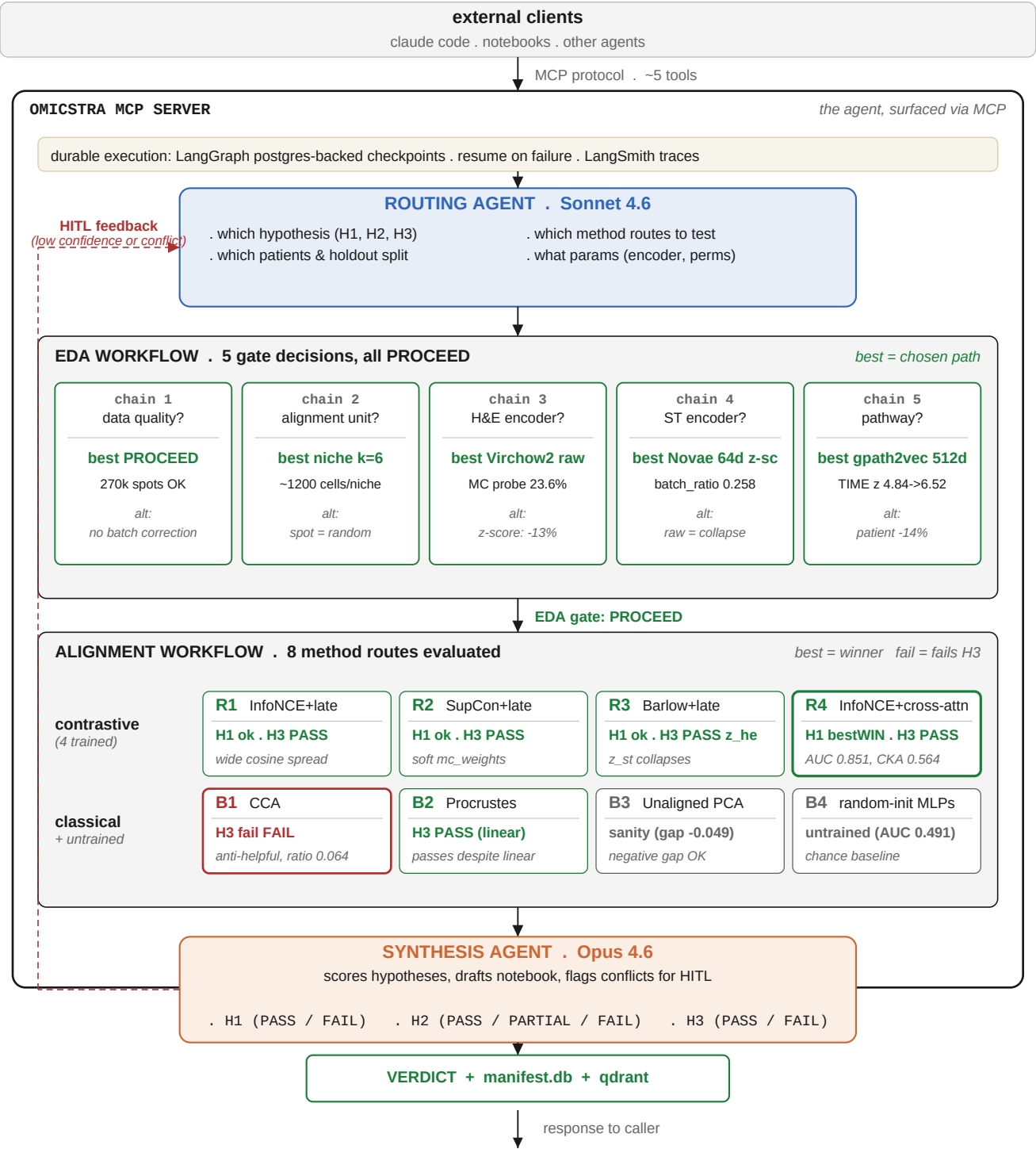
scientific brief . companion to `current_report.md` and `tnbc92_summary.pdf`

this brief describes the H1, H2, H3 evaluation framework for the tncb-92 cross-modal alignment work in the omicstra package: what each hypothesis measures, how the metrics are defined, what the eight method routes are, what each test found, and which method is recommended for which class of downstream question. companion to the method-by-question routing matrix used by the alignment-agent. intended for computational biologists, research software developers, and PIs evaluating the system.

omicstra: routing decisions on every path

status: planned architecture (v0 spec) - agent layer not yet built

the EDA chains (page 4) and alignment runs (page 5) on which all H1/H2/H3 results are based are **complete and validated**. the MCP server, LangGraph orchestration, Sonnet routing, Opus synthesis, and HITL feedback layer shown below are the **target system design** (ResearchHub Foundation grant deliverable, in scoping). src/ is empty at time of writing; scientific results in this brief come from manual orchestration of the underlying scripts (align.py, eval.py, eval_alignment_biology.py).



three questions, three label requirements

why three hypotheses, not one

cross-modal alignment between H&E and spatial transcriptomics has three properties that need to be tested independently: whether the two modalities couple (H1), whether the shared space preserves biological structure (H2), and whether the embedding amplifies biology relative to confounders (H3). a method can pass one and fail another. **CCA passes H1 weakly, fails H2 by patient leakage, and fails H3 by patient amplification**, and yet on a single ARI metric it would have looked like the best biology-preserving method in the grid. ranking methods on a single metric collapses this structure and can pick the wrong winner.

each hypothesis answers a different scientific question with different label requirements. the falsification structure across the three families is what makes the verdict on any single run survive scrutiny rather than depending on one favored metric.

H	scientific question	what passing tells you	label requirement
H1	does H&E predict its matched ST niche better than chance after alignment?	cross-modal coupling, ex. the model finds shared signal at all	matched (H&E, ST) pairs only; no biology labels needed
H2	does the shared embedding preserve biological compartment structure?	structure preservation, ex. manifold is organized by biology, not patient	tissue compartments + niche-level biology (Wang mc_megacluster, NMF per-spot)
H3	do the shared embedding directions correlate with named pathways, with cross-patient generalization?	pathway-level interpretability that transfers across patients	5 named Reactome pathway AUCell scores (niche-level) + Bareche TIME/MC labels (robustness)

interlock structure

passing all three is the load-bearing claim. a method that passes all three is doing genuine cross-modal alignment that preserves biology and disentangles confounders. a method that fails any one has a specific identifiable failure mode:

H1	H2	H3	interpretation	example
pass	fail	fail	couples but reads patient identity as biology	B1 (CCA)
pass	partial	win cross-patient	couples and generalizes; niche-level cluster geometry suffers under tight manifold	R4
pass	win cluster	fail cross-patient	couples and preserves niche-level biology, but pathway encoding is patient-specific	R1, R6
pass	win	win	genuine cross-modal alignment with BOTH niche-resolution AND cross-patient generalization	target end-state (none of 8 runs achieves both)
fail	pass	pass	doesn't couple modalities; H2/H3 signal is unimodal	would invalidate cross-modal claim

what we chose, what we ruled out

before any alignment training, 5 sequential gate decisions checked whether the dataset is usable, what the right alignment unit is, and which encoders to commit to. all 5 gates passed. this section walks through each chosen path and the alternatives that were rejected with their measured penalty.

chain	question	chosen	evidence	rejected alternative
1	dataset usable?	PROCEED	MKI67 0.5, CD3D 0.15 detected; Moran's I 0.6 (COL1A1); batch silhouette < 0.1	no batch correction needed; ESR1/PGR ~0 confirms TNBC
2	alignment unit?	niche k=6 (~1200 cells)	spot-level cross-patient R@1 = 0.002 (random); within-patient R@1 = 0.136 (1.6x amplification)	spot-level alignment (empirically infeasible at this cohort size)
3	H&E encoder?	Virchow2 raw	MC probe 23.6%, morph probe 57.1%, batch ratio 1.089, PathoROB RI=0.88 (highest of 20 FMs tested)	UNI2 raw (probe 2.2%); Virchow2 z-scored (probe drops ~13%)
4	ST encoder?	Novae 64d z-scored per subarray	batch_ratio 0.258 (biology-dom); effective rank 11 of 64; 19 subarrays excluded (<512 spots)	Novae raw (cross-subarray incomparable; silhouette -0.246 with mean shift)
5	pathway embedding?	gpath2vec 512d	TIME bio z 4.84 -> 6.52 (1.35x); patient z 34.30 -> 29.63 (-14%); rho 0.955 vs raw EA matrix	raw enrichment scores (no patient compression, asymmetric smoothing absent)

the load-bearing decisions, expanded

why niche, not spot. the Wang et al. cohort uses original ST with 100um spots (~200 cells each), not 10x Visium. at spot resolution, cross-patient retrieval is empirically random (R@1 = 0.002 across ~270k spots, chance ~ 4e-6). aggregating to niches of k=6 spatial neighbors (~1200 cells per niche) captures the tumor microenvironment architecture that biology labels (TIME, MC subtypes) actually describe at the tissue-region scale.

why Virchow2 RAW. z-scoring per subarray is the standard preprocessing for cross-batch normalization. it hurt Virchow2's MC probe by 13 percentage points (23.6% -> 10.4%), because Virchow2 already encodes morphological structure cross-subarray and the per-subarray mean is biologically meaningful. PathoROB RI=0.88 (Komen et al. 2025) is the highest reported reproducibility index across 20 H&E foundation models, validated independently. encoder-specific preprocessing is the lesson: H&E and ST do not share the same input pipeline.

why Novae 64d Z-SCORED. opposite finding. Novae was trained on Xenium subcellular ST; our cohort is 100um original-ST. cross-platform mismatch is real. per-subarray z-scoring recovers cross-subarray comparability (silhouette improves from -0.246 to -0.135). the residual geometric inconsistency is genuine cross-platform drift, not removable. 19 subarrays excluded for < 512 spots (Novae's architectural floor).

why gpath2vec AND AUCell-5. gpath2vec (286k niches x 512d) validated as ST input feature: biology preserved (TIME z 4.84 -> 6.52), patient compressed (-14%). The proposal-spec H3 test (pages 17-22) uses AUCell scores at niche level for the 5 named Reactome pathways instead of gpath2vec full-dim. AUCell-5 is the pathway side of the per-pathway CCA test, with 100% test-set coverage including ECM (TF-filter concern resolved). R6 ablation confirms z_he biology does NOT depend on gpath2vec being in ST, so the inclusion is not circular.

niche-level label addition: mc_megacluster (Wang per-spot 14-class NMF discrete label) joined into niche-join parquets (94% coverage). niche-level biological label that the original EDA chain treated only as mc_weights (continuous, supervision-only). Used in H2 part A-MC.

8 method routes, what each tests

after EDA gates passed, 8 alignment configurations were run on the same 260 matched subarrays, 85/15 patient-stratified split, seed=42: 4 contrastive variants (R1-R4) + 3 classical baselines (B1-B3) + 1 ablation (R6). this section explains what each route tests; subsequent sections (H1, H2, H3) score them.

alignment architecture (two-stream)

every run aligns two streams via contrastive loss on matched (H&E, ST) niche pairs. **H&E stream**: virchow2_niche (1280d, mean-pooled over self + 6 spatial neighbors). **ST stream**: novae_niche (64d, z-scored per subarray) concatenated with gpath2vec_niche (512d) = 576d. each stream goes through an MLP (1280 -> 512 for H&E, 576 -> 512 for ST). loss is computed on cosine similarity between matched pairs vs cross-patient mismatched pairs in each batch.

the 8 routes

run	loss	fusion	ST input	what this route tests
R1	InfoNCE	late MLPs	novae+gpath	baseline contrastive: does softmax over cross-patient negatives align modalities?
R2	SupCon	late MLPs	novae+gpath	does mc_weights soft supervision beat unsupervised InfoNCE?
R3	Barlow	late MLPs	novae+gpath	does redundancy reduction work without softmax negatives?
R4	InfoNCE	cross-attention	novae+gpath	does ST query attending H&E tile tokens beat mean-pooling?
R6	InfoNCE	late MLPs	novae only	does z_he biology depend on gpath2vec being in ST? (ablation)
B1	CCA (closed-form)	linear correlation	novae+gpath	classical: does linear correlation maximization work cross-modally?
B2	Procrustes	orthogonal rotation	novae+gpath	does orthogonal alignment without reweighting work?
B3	unaligned PCA	no coupling	novae+gpath	sanity check: independent PCAs should NOT align (negative gap)

R4 cross-attention deviation

R4 differs only on the H&E side. instead of mean-pooled virchow2_niche, R4 uses 7 raw virchow2_cell tile tokens (center spot + 6 neighbors), with cross-attention from the ST query selecting which tiles matter. ST side identical to R1. this is the architectural reason R4 lands in a tight ~3-effective-dim manifold while R1 spreads across ~5 dims: cross-attention compresses the H&E representation to the subset of tiles that ST queries actually attend to.

which method wins which hypothesis

this is the at-a-glance scorecard across all 8 method routes. detailed explanations of why each method passes or fails each hypothesis follow in the H1, H2, H3 sections. **headline findings:** R4 (cross-attention) wins H1 retrieval AND H3 cross-patient pathway generalization. R1/R6 win H2 niche-level biology preservation. B1 (CCA) is anti-helpful on H3-extended. B2 (Procrustes) passes H3-extended despite being linear, falsifying the original “linear methods fail” prediction.

run	H1 (alignment)	H2 (structure)	H3 (pathway interpretability)
R4	best WIN AUC 0.851, CKA 0.564	Part A: low (patient suppression). Part A-MC: low KMeans (geometry bias) but LP parity 0.230 . Part B: 0/6 contrasts.	Part A: pass. best Part B WIN cross-patient (15/15, z 3.8-5.7) . DAG decomposition: distinct sub-programs (BTLA, IFN, integrin, collagen).
R1	PASS AUC 0.741, CKA 0.242	Part A: 0.250 archetype. Part A-MC: WIN KMeans 0.231, LP 0.260 . Part B: 2/6 (TLS only).	Part A: pass. Part B FAIL (5/15) - pathway encoding is patient-specific.
R2	PASS AUC 0.707, gap 0.043	Similar to R1, weaker LP 0.229. Part B: 2/6.	Part A: pass. Part B FAIL (5/15).
R3	PASS AUC 0.646, CKA 0.252	Part A-MC LP lowest 0.182. Part B: 3/6 (inverted geometry).	Part A: pass. Part B FAIL (5/15).
R6	PASS AUC 0.733	Part A-MC: WIN KMeans 0.251 , LP 0.235. Part B: 2/6 (TLS only).	Part A: pass. Part B partial (11/15: z_st 5/5, z_he 1/5 - counter-intuitive).
B1	weak AUC 0.541, CKA 0.086	Part A: 0.307 (patient leakage win). Part A-MC: drops 34% on shift.	Part A: pass. Part B: 14/15. H3-extended fail FAIL (ratio 0.064 < raw floor 0.129).
B2	PASS classical AUC 0.703, CKA 0.124	Rotation-equivalent to B3 on H2. Part A-MC LP 0.233.	Part A: ties B3 cumulative win. Part B: 14/15. H3-ext PASS (linear).
B3	fails AUC 0.443, gap -0.049	Part A-MC LP 0.237. rotation-equivalent to B2.	Part A: ties B2 (rotation-eq). Part B: 12/15.

how to read this scorecard

H1 column. AUC and CKA-after measure how well matched (H&E, ST) niche pairs cluster vs unmatched. all contrastive methods (R1-R4) outperform all classical baselines (B1-B3) on both metrics with no overlap. R4 leads by a wide margin, and a rank-matched control rejects the manifold-collapse explanation. detailed in section H1 (pages 7-9).

H2 column (substantially updated since v1). the original H2 metric (ARI on 9 archetypes) was found to be **mechanically a patient classification test** – archetype labels are patient-level pseudobulk (verified: every niche in a patient inherits one label). the proper niche-level test uses Wang’s mc_megac_luster (per-spot 14-class NMF), with three metric-independent diagnostics (linear probe, kNN purity, rank-matched KMeans) added when R4 came out lowest on KMeans. final reading: R1/R6 win KMeans MC ARI; R4 passes the patient-stratified linear probe at parity. Part B compartment cosine: 1/6 contrasts supported (TLS-vs-TIL only) due to annotation-purity confound. detailed in

section H2 (pages 10-13).

H3 column. the proposal-spec test is per-pathway CCA on 5 named Reactome parent subtrees, with permutation null and BH-FDR. **two distinct tests: Part A (fit-on-all)** - every run passes BH-FDR; the proposal hypothesis is supported universally but the test is saturated by 512-d projection capacity onto 1-d pathway scores. **Part B (cross-patient sub-split)** - R4 dominant (15/15), classical baselines strong (12-14/15), late-fusion contrastive fail (5/15). a DAG decomposition pushed the test to all 395 sub-pathway nodes (9480 cells); R4 still leads, classical baselines stay competitive, R1/R6 fall near floor. the bio-vs-patient z-test on Bareche TIME / MC labels (v1's H3) is preserved as **H3-extended robustness analysis**. detailed in section H3 (pages 17-22).

alignment quality

- Q does learned contrastive alignment improve cross-modal H&E to ST matching over classical baselines? this is the foundational test. if the two modalities don't couple at all in the shared space, none of the downstream interpretability claims hold.

theoretical motivation

H&E captures tissue morphology; ST captures gene expression. they are physically locked to the same tissue, so they should share information about underlying biology. a successful alignment finds a low-dimensional space where matched (H&E, ST) pairs are closer than unmatched pairs.

classical methods (CCA, Procrustes) use linear projections; contrastive methods (InfoNCE, SupCon, Barlow) use non-linear projections trained with the explicit objective of pulling matched pairs together and pushing unmatched apart, with cross-patient negatives in each batch. H1 asks whether the non-linear training buys anything over the linear baselines.

method, mathematically: what the H1 metrics actually compute

setup. for every test niche i , the model produces two unit-norm vectors: z_{he_i} (from H&E, 512-d) and z_{st_i} (from ST, 512-d). matched pairs share the same i ; mismatched pairs come from different niches.

R@K separates by ranking. for each query z_{he_i} , score every gallery z_{st_j} by cosine similarity, sort, and check whether the matched $j=i$ appears in the top K. $R@K$ = fraction of queries where it does. ex. $R@1 = 0.001$ means 0.1% of queries land their match in position 1.

AUC separates by ordering. probability that a random matched pair has higher cosine than a random mismatched pair. $AUC = 0.5$ is chance; $AUC = 1.0$ is perfect ordering. scale-invariant, so it doesn't care if cosines are clustered tight or spread wide as long as matched > mismatched.

alignment gap separates by absolute distance. $\text{mean}(\text{matched cosines}) - \text{mean}(\text{mismatched cosines})$. positive means matched pairs are systematically closer in cosine units. ex. $R4 \text{ gap} = 0.195$ means matched pairs are on average 0.195 cosine units closer than random pairs.

CKA-after separates by space alignment, not pair alignment. compute Gram matrices $K_{HE} = Z_{HE} @ Z_{HE}.T$ and $K_{ST} = Z_{ST} @ Z_{ST}.T$ (these are n -by- n matrices of pairwise cosines within each modality). center them, then take the normalized inner product: $\text{HSIC}(K_{HE}, K_{ST}) / \sqrt{\text{HSIC}(K_{HE}, K_{HE}) \cdot \text{HSIC}(K_{ST}, K_{ST})}$. tests whether the two spaces have similar pairwise structure, even if individual pairs aren't matched.

the falsifier for collapse-driven CKA inflation

a tighter manifold mechanically scores higher CKA against any partner, because dimension collapse trivially reduces between-space mismatch. so a high CKA could mean (a) genuine alignment or (b) one space collapsed to fewer dimensions. the falsifier: project R1 to its top-3 principal components (matching R4's effective rank), recompute CKA, compare to R4's full-dim CKA. R4@rank3 is the sanity check. should approximately match R4 full-dim since R4 already lives in ~3 dims.

rank projection, mathematically

given Z_{HE} shape (n, 512), compute its SVD: $Z_{HE} = U S V.T$

rank-3 projection: $Z_{HE_rank3} = U[:, :3] @ \text{diag}(S[:3]) @ V.T[:3, :]$

this is the best rank-3 approximation of Z_{HE} in Frobenius norm. if R1 already lives mostly in its top-3 PCs, $Z_{HE_rank3} \sim Z_{HE}$ and CKA barely moves. if R1 spreads its variance across more dimensions, Z_{HE_rank3} loses information and CKA drops.

interpretation: if R1@rank3 reaches R4 full-dim CKA, R4's H1 advantage is just compression. if R1@rank3 stays at R1 full-dim CKA, R4 is doing real architectural work that compression alone cannot reproduce.

run	full-dim CKA	rank-3 CKA	gap
R1	0.242	0.234	-0.008
R4 (sanity)	0.564	0.564	-0.0003
R1@rank3 vs R4 full	-	-	-0.331

R1@rank3 (0.234) does *not* reach R4 full (0.564). the gap of -0.331 places R4 firmly in the “real architectural advantage” bucket. R4's cross-attention is doing actual cross-modal work, not just compressing the manifold to inflate CKA.

results across all 9 runs

run	method	R@1	R@5	AUC	CKA-after	gap
R4	InfoNCE + cross-attn	0.0002	0.0014	0.851	0.564	0.195
R1	InfoNCE + late	0.0003	0.0010	0.741	0.242	0.159
R6	InfoNCE + late (novae-only)	0.0002	0.0007	0.733	0.225	0.155
R2	SupCon + late	0.0002	0.0009	0.707	0.120	0.043
R3	Barlow + late	0.0001	0.0007	0.646	0.252	0.125
B2	Procrustes	0.0002	0.0008	0.703	0.124	0.156
B1	CCA	0.0000	0.0002	0.541	0.086	0.007
B4	random-init MLPs (untrained)	0.0000	0.0001	0.491	0.110	-0.001
B3	Unaligned PCA	0.0000	0.0000	0.443	0.124	-0.049

what each result means

family separation is clean. all four contrastive runs (R1-R4) and the contrastive ablation R6 outperform all three classical baselines (B1-B3) and the untrained baseline (B4) on AUC and CKA-after. AUC range: contrastive 0.646-0.851, classical 0.443-0.703, untrained 0.491. CKA range: contrastive 0.120-0.564, classical 0.086-0.124, untrained 0.110.

R4 leads both axes by a wide margin. AUC 0.851 (vs R1 0.741), CKA 0.564 (vs R1 0.242, vs R3 0.252). its tight ~3-effective-dim manifold is the regime where cross-attention earns its keep. the ST query selectively attends over 7 H&E tile tokens (center spot + 6 neighbors) instead of mean-pooling, capturing morphology-relevant tile selection that fixed pooling cannot.

R@K is low in absolute terms across all runs (~10⁻⁴ at R@1). this is consistent with niche-level cross-patient retrieval being empirically near-impossible at this cohort size. for context: even R4 (the H1 winner with AUC 0.851) achieves R@1 = 0.0002 on the held-out test set of ~40k niches, and B3 unaligned PCA achieves R@1 = 0.0000. the chance floor is ~2.5e-5 at this gallery size. the four-orders-of-magnitude span in AUC (0.443 to 0.851) is what separates methods, not absolute retrieval thresholds. our spot-level alignment pilot (chain 2, EDA) confirmed the same finding: cross-patient spot-level R@1 = 0.002 (random) vs within-patient R@1 = 0.136. the proposal's H1 evaluation grades on relative-improvement-over-baseline.

B3 unaligned PCA produces negative alignment gap (-0.049). this is the correct sanity check: independent PCAs of two modalities should not align cross-modally in cosine space. if B3 had positive gap, the experimental setup would be broken.

B4 random-init MLPs sit at chance (AUC 0.491, gap -0.001). this is the proposal-strict non-contrastive late-fusion baseline: the R1 architecture with Xavier-initialized weights and no training. B4 isolates what the contrastive training contributes vs what the architecture alone contributes. since B4 is essentially at chance, the contrastive learning is doing the bulk of the work in R1's AUC 0.741 result, not the late-fusion MLP architecture itself.

B1 CCA collapses on AUC (0.541). whitened canonical correlation does not survive L2-normalization into cosine space well. CCA maximizes correlation in the whitened pre-normalization space, and that signal is largely killed by the unit-norm projection that puts everything on the cosine sphere.

R3 (Barlow) and R2 (SupCon) sit between R1 and the classical baselines on AUC. Barlow's redundancy-reduction loss does not explicitly push matched pairs together (only off-diagonal cross-correlation toward zero), so its alignment gap is smaller than InfoNCE's. SupCon's current numbers are pessimistic because mc_weights soft targets had uniform-imputed NaNs in ~half of niches; a NaN-cleaned rerun is on the queue.

VERDICT

PASS. learned contrastive alignment captures cross-modal structure that classical methods cannot. R4's architectural advantage is real, not collapse-driven. the binary "contrastive > classical" claim holds on both AUC and CKA without overlap between families.

structure preservation

- Q does the shared embedding preserve biological compartment structure beyond either modality alone? if alignment compresses everything onto a generic shared axis without preserving the per-niche biological program, the embedding has cross-modal coupling (H1) but no interpretability. every niche looks alike.

theoretical motivation

a useful alignment should organize niches in the shared space by *biological identity* (this is a tumor niche, that is a stroma niche, that is a TLS niche) rather than by patient identity. the pre-registered metric was Adjusted Rand Index (ARI) on 9 archetype labels: a clustering-vs-labels agreement test that asks “does k-means on the shared space recover the archetype structure?”

this metric has a hidden confound that became apparent only after the H3 disentanglement test was run on the same runs. the methodology evolved in response.

method, mathematically: what ARI computes and why it failed here

step 1. run k-means on z_{he} with $k=9$ (matching the 9 archetype labels). this produces a partition C of niches into 9 clusters.

step 2. compare C to the ground-truth archetype partition T . ARI counts pairs of niches that are in the same cluster in *both* C and T , vs pairs that disagree. correction-for-chance: subtract expected agreement under random labeling, divide by max possible agreement minus expected.

formula. $\text{ARI} = [\sum_{ij} (n_{ij} \text{ choose } 2) - E] / [(1/2)(\sum_i (a_i \text{ choose } 2) + \sum_j (b_j \text{ choose } 2)) - E]$, where n_{ij} is the count of niches in cluster i of C and class j of T ; a_i, b_j are row/column sums; E is the expectation under the hypergeometric null. ARI = 0 is chance; ARI = 1 is perfect agreement.

why it failed here. the 9 archetype labels are defined per patient. each patient gets their own archetype assignment from a separate clustering pass. so “niches with the same archetype” tend to be from the same patient. ex. a method that clusters niches by patient identity gets a high ARI on archetypes *even if it has zero biology signal*. ARI on per-patient labels measures patient leakage, not biology preservation.

ARI tracks patient leakage, not biology

run	z_he ARI	reading
B1 (CCA)	0.307	leakage, patient z amplified 3.3x
B2 (Procrustes)	0.298	preserves PCA + leakage
B3 (unaligned)	0.298	unaligned + leakage
R6 (novae-only)	0.254	patient-saturated z_st
R1 (InfoNCE+late)	0.250	late fusion
R2 (SupCon+late)	0.204	SupCon
R3 (Barlow+late)	0.129	Barlow
R4 (InfoNCE+cross-attn)	0.059	cross-attn (compressed manifold)

the ARI ranking exactly tracks patient-axis preservation: B1 wins ARI *and* amplifies patient z 3.3x on the same axis. R4 has the lowest ARI *and* the strongest patient suppression. this is direct evidence that ARI here reads patient leakage, not biology.

rotation-invariance footnote: B2 = B3 mathematically

B2 and B3 are identical to four decimal places on every H2 metric. this is mathematically guaranteed, not a coincidence.

why B2 = B3 on H2 but not on H1

H2 metrics are rotation-invariant. ARI and cosine silhouette depend only on pairwise cosines within one space. if $Z_{B3} = Z_{B2} R$ for some orthogonal matrix R (which is exactly what Procrustes adds on top of unaligned PCA), then $\cos(Z_{B3_i}, Z_{B3_j}) = (Z_{B2_i} R) \cdot (R^T Z_{B2_j}) = Z_{B2_i} \cdot Z_{B2_j} = \cos(Z_{B2_i}, Z_{B2_j})$. all pairwise cosines preserved, all cluster assignments preserved, all silhouettes preserved.

H1 metrics are NOT rotation-invariant. AUC and alignment_gap depend on cross-modal cosines: $\cos(z_{he_i}, z_{st_i})$. Procrustes rotates one modality to align with the other, which changes these cross-modal cosines. B2 AUC = 0.703 (decent), B3 AUC = 0.443 (sub-chance). the rotation matters for cross-modal pairing even though it doesn't matter for within-space clustering.

practical consequence. the H2 ranking gap between B2 and B3 is exactly zero by construction. any reported H2 difference would be a numerical artifact, not science.

KMeans MC ARI and metric-independent diagnostics

after the label audit, `mc_labels.megacluster` (Wang per-spot 14-class NMF discrete label) was joined into the niche-join parquets. **this is the proper niche-level biological clustering target**. KMeans(k=14) plus a patient-stratified linear probe (11/3 within the 14 test patients), plus kNN purity and rank-matched KMeans, form the diagnostic chain.

the initial K-means MC ARI result

run	ARI (MC, z_he)	reading
R6 (novae-only ST)	0.251	best niche-level biology preservation
R1 (InfoNCE+late)	0.231	runner-up
B2 / B3	0.205	classical, lost 31% on resolution shift
B1 (CCA)	0.202	classical, lost 34% on shift (patient leakage gone)
R2 (SupCon+late)	0.174	mid
raw_he	0.174	lost 39% on shift - was patient identity
R3 (Barlow+late)	0.166	low
R4 (InfoNCE+cross-attn)	0.098	lowest

Classical baselines and `raw_he` lose 31-39% of ARI under the resolution shift from patient-level to niche-level labels – **quantitative confirmation of the patient-leakage diagnosis on Part A**. R4 came out lowest, inconsistent with its H1 dominance and clean H3-extended biology amplification (TIME z = 10.07 vs `raw_he` 4.79). This triggered commit 2.6's metric-independent diagnostics.

metric-independent diagnostic refinement

three pre-registered diagnostics, each metric-independent in a different way:

diagnostic 1 – kNN purity (k=5/10/25): niches' k-NN in `z_he` share `mc_megacluster`? All runs >> chance (>0.5 at k=10 vs chance 0.071), but values are dominated by within-patient niche similarity (intra-patient niches share MC because MCs are spatially smooth within tumors). High purity at baseline = within-patient clustering. R4's lower value (0.65 vs baselines 0.80) is consistent with R4's known patient suppression, not biology loss. **kNN purity without patient stratification is patient-confounded; not interpretable as a biology metric.**

the metric-independent verdict on R4

diagnostic 2 – linear probe (LogReg, 14-class, patient-stratified): 11/3 patient sub-split within 14 test patients. rank1a Virchow2 niche baseline: 21.4% (3x chance).

run	LP MC accuracy	reading
R1 (InfoNCE+late)	0.260	highest
raw_he	0.239	raw modality baseline
B3 (Unaligned PCA)	0.237	-
R6 (novae-only)	0.235	-
B2 (Procrustes)	0.233	-
R4 (InfoNCE+cross-attn)	0.230	best parity with R1/raw_he
R2 (SupCon+late)	0.229	-
B1 (CCA)	0.210	-
R3 (Barlow+late)	0.182	lowest

load-bearing finding: R4 (0.230) sits within 3 percentage points of R1 (0.260) and raw_he (0.239). **Geometry-agnostic test shows R4 encodes MC at parity with the K-means winners.** R4 is NOT biology-less.

diagnostic 3 – rank-matched KMeans (project z_he to top-3 PCs):

run	full ARI	rank-3 ARI	reading
R6	0.251	0.185	-
R1	0.231	0.193	16% loss to dim reduction
B1	0.202	0.084	-
R4	0.098	0.095	sanity: R4 lives in ~3 dims

even at matched rank, R1@rank3 (0.193) is 2x R4 full (0.098). **R4's gap is roughly half geometric (dim-collapse) and half encoding-shape (linearly-separable but non-globular).** KMeans assumes globular cluster regions; R4 stores MC along linearly-separable axes. LogReg only needs linear separation, which R4 has.

the resolved verdict on R4 (replaces initial 2.5)

measurement	finding	what it says about R4
KMeans MC ARI (full or rank-3)	R4 0.098, R1/R6 0.19-0.25	measurement artifact of globular-cluster assumption
kNN purity (k=5)	R4 0.652 vs baselines 0.80	patient-confound: low value reflects R4's patient suppression, not biology loss
linear probe (patient-stratified)	R4 0.230 , R1 0.260, raw_he 0.239	biology at parity with all other runs
H1 AUC	R4 0.851 (best)	cross-modal alignment intact
H3-ext z_he TIME bio z	R4 10.07 vs raw_he 4.79 (2.1x amp)	biology amplified, just compressed

R4 cross-attention preserves niche-level biology at parity with R1/R6/raw_he by the only metric-independent test (patient-stratified linear probe). R4's KMeans MC ARI of 0.098 was a **K-means / globular-cluster bias against tight manifolds**, not a biology preservation failure.

the proposal's directional prediction

The proposal's directional prediction: morphologically distinct compartments (tumor, TLS, necrosis) show *higher* matched-pair cosine $\cos(z_{he}[i], z_{st}[i])$ than ambiguous compartments (high-TIL stroma, low-TIL stroma). 6 named contrasts per run: 3 distinct $\times$ 2 ambiguous. Welch z-test (unpooled variance). Lymphoid nodule is the closest proxy for TLS in Wang annotations.

falsifier-failure result

contrast	R1	R2	R3	R4	R6	B1	B2	B3
Tumor vs hiTIL	fail	fail	fail	fail	fail	pass ns	fail	fail
Tumor vs loTIL	fail	fail	fail	fail	fail	pass ns	fail	fail
TLS vs hiTIL	pass z=+3.2	pass	fail	fail	pass	pass ns	fail	fail
TLS vs loTIL	pass z=+5.3	pass	fail	fail	pass	pass ns	fail	pass
Necrosis vs hiTIL	fail	fail	pass	fail	fail	pass ns	pass	fail
Necrosis vs loTIL	fail	fail	pass	fail	fail	pass ns	fail	fail
pass count	2/6	2/6	3/6	0/6	2/6	5/6 ns	1/6	2/6

Only **TLS vs TIL-stroma** supported across contrastive runs (R1/R2/R6). Tumor and Necrosis contrasts INVERT with statistical decisiveness ($z \leq -2.8$). B1's 5/6 directionally-correct contrasts all non-significant – uniform low-effect cosines, no real per-compartment differentiation.

mechanism hypothesis (not pre-registered)

The failure is **annotation- and sample-size-driven, not a refutation of cross-modal alignment quality**. High TIL stroma (n=105) and Low TIL stroma (n=354) are small, well-curated compartments – internally homogeneous biology. Tumor (n=4384) and Necrosis (n=1178) are large, heterogeneous compartments – TNBC subtype variation alone produces within-compartment cross-modal decorrelation. The matched-pair cosine averages over more biologically diverse niches. Lymphoid nodule (n=44) shows the predicted direction in R1/R2/R6 because it is morphologically uniquely distinct (dense lymphocyte aggregation), and its low n keeps within-compartment biology homogeneous. A re-run on the 90-subarray annotated subset with the full 18-class compartment vocabulary (Tumor central vs Tumor edge, Tumor stroma rich, etc.) may recover the predicted direction; the current 12-class collapsed vocabulary appears too coarse.

the H2 picture, three tests, three readings

Part A archetype ARI: **FAIL** as a biology test. Labels are patient-level. Metric is mechanically a 9-class patient classification.

Part A-MC niche-level MC ARI: R1/R6 win on KMeans (0.231-0.251). R4 fails KMeans (0.098) due to manifold geometry. Classical baselines lose 31-37% on resolution shift confirming patient-leakage diagnosis.

Part A-MC diagnostics: R4 passes **linear probe at parity** with R1/R6/raw_he. K-means handicap is geometry-bias against tight manifolds. Rank-matched KMeans decomposes R4's deficit into roughly half geometric, half encoding-shape (linearly-separable but non-globular). LogReg parity confirms encoding-shape half is not biology absence.

Part B compartment cosine: 1/6 contrasts supported (TLS vs TIL only, in R1/R2/R6). Tumor and Necrosis contrasts inverted due to annotation-purity confound, not biology absence.

VERDICT

PARTIAL -> fusion-axis trade-off. H2 is a fusion-axis trade-off, not winner-take-all. Late-fusion contrastive (R1, R6): best on niche-level MC clustering by both K-means and linear probe; preserve niche-resolution biology with distinguishable regions. Cross-attention bridge (R4): matches R1/R6 on linear MC decodability but lower K-means due to manifold tightness; trades globular cluster geometry for cross-modal alignment quality (H1) and cross-patient pathway transfer (H3). Classical baselines (B1, B2, B3): Part A archetype 'wins' were patient leakage; on the proper niche-level test they lose 31-37% of ARI.

per-pathway CCA on 5 named Reactome pathways

Q (proposal-s pec) do gpath2vec Reactome embeddings for 5 named pathways (TGF-β R-HSA-170834, Immune R-HSA-168256, ECM R-HSA-1474244, Cell Cycle R-HSA-1640170, PCD R-HSA-5357801) correlate with specific directions in the shared latent space via CCA, indicating preservation of interpretable biological signal?

pathway **signal:** niche-level AUCell scores from
data/embeddings/biological_signals/niche_aucell_5targets.parquet (286k niches × 5 pathways, 100% test-set coverage; ECM present, TF-filter concern resolved on this artifact).

method: univariate CCA via closed-form lstsq (~15 lines). pathway side is 1-d, so univariate CCA reduces to pearsonr(z_proj, y) where z_proj = Z @ w and w = lstsq(Z_centered, y_centered)/||w||. Two CCA variants:

Part A (option B in code): fit on all test niches – proposal's literal test. **Part B (option A in code):** cross-patient sub-split (fit on 11 train patients within test set, eval on 3 held-out). **permutation null:** 1000 perms × 120 cells per option, BH-FDR.

Part A – fit-on-all CCA (proposal's literal test)

null mean = 0.106 across every cell (= sqrt(d/n) = sqrt(512/45661), the theoretical OLS overfit floor). Observed 0.70-0.90, z-scores 180-245. **all 120 cells BH-FDR < 0.001.**

run	cumulative corr (5 pathways x 3 views)
B2 / B3	4.242 (rotation-equivalent decisive win, +0.143 over R1)
R1	4.099
R6	4.093
B1	4.092
R2	4.079
R3	4.021
R4	3.943

every run encodes pathway-aligned information that survives null calibration. the proposal's H3 hypothesis is supported for every run. But the test is saturated by 512-d projection capacity onto a 1-d target – narrow spread, no architectural ranking power.

sub-finding: B1 CCA loses cumulative to B2/B3

CCA (B1) optimizes cumulative canonical correlation *by construction*. Losing to rotation-equivalent B2/B3 (Procrustes / unaligned PCA) by 0.15 cumulative is itself diagnostic. Most likely B1's whitening + projection compresses variance that B2/B3 preserve through orthogonal rotation. Logged as a CCA-baseline pathology; not load-bearing for the H3 verdict.

the discriminating test

CCA fit on 11/14 test patients, evaluated on 3 held-out patients. BH-FDR < 0.05 over 120 cells.

run	sig cells / 15	z_he z-scores (TGF- β / Immune / ECM / CC / PCD)
R4	15 / 15	4.5 / 4.6 / 5.7 / 4.2 / 3.8
B1	14 / 15	2.8 / 3.5 / 4.4 / 2.4 / 1.6
B2	14 / 15	3.0 / 3.5 / 4.8 / 2.5 / 1.3
B3	12 / 15	3.0 / 3.5 / 4.8 / 2.5 / 1.3
R6	11 / 15	z_st 5/5 + z_mean 5/5 driven; z_he 1/5 (only ECM) – counter-intuitive
R1	5 / 15	1.2 / 1.3 / -0.3 / 1.5 / 1.4
R2	5 / 15	-1.6 / -0.6 / 1.1 / -0.9 / -0.7
R3	5 / 15	-0.6 / -1.4 / -1.7 / 1.3 / -0.0

R4 dominates cross-patient generalization (15/15 significant, z-scores 3.8-5.7). R1/R2/R3 barely above the 5% FDR baseline.

specificity diagnostic – how R4 wins

z_he off-diagonal |mean| of pairwise canonical-direction cosines across 5 pathways:

run	off-diag mean	max	reading at 5-parent resolution
R4	0.819	0.970	highly correlated at parent level
R1	0.481	0.771	mid
R6	0.446	0.733	mid
B1	0.411	0.742	mid
R2	0.407	0.639	distinct
R3	0.390	0.723	most distinct
B2 / B3	0.376	0.684	distinct

at 5-parent resolution R4's pathway directions are highly correlated. the natural reading is that R4's tight ~3-effective-dim manifold cannot afford 5 orthogonal pathway directions, so the 5 (already-correlated) Reactome parent subtrees collapse onto roughly one biology axis. that axis transfers cleanly cross-patient (15/15).

this reading is partially correct but incomplete. the 5 parent subtrees share large amounts of gene content (Immune R-HSA-168256 has 219 nodes / 145 leaves; ECM R-HSA-1474244 has 20 nodes / 13 leaves; the unions overlap heavily). AUCell scores on parent unions are inherently correlated, and any method's CCA directions on correlated targets will also be correlated. the DAG decomposition on the next page (testing R4 against 395 sub-pathways individually) shows that R4 encodes biologically distinct sub-programs (BTLA checkpoint, IFN-alpha/beta, integrin adhesion, collagen biosynthesis, collagen degradation, NFAT signaling) and that the 0.819 off-diagonal is a coarse-aggregation artifact of the 5-parent test, not a property of R4's encoding.

the asymmetric stream finding + Bareche-label check

R6 counter-intuitive direction (verified from parquet)

R6 dropped gpath2vec from the ST stream, so we would expect z_{st} to be the *weakest* pathway-correlated view (no explicit pathway anchor in input). Observed: opposite – z_{st} is 5/5 significant cross-patient (z-scores 2.3-5.0), z_{he} is 1/5.

plausible mechanism: contrastive training with no pathway anchor on ST forces z_{he} to compress toward z_{st} 's geometry through the InfoNCE objective, eroding z_{he} 's native pathway alignment. z_{st} alone retains implicit biology correlation through Novae's spatial-neighborhood GAT structure even without an explicit pathway feature. **R6b mirror ablation** (gpath2vec-only ST, novae dropped) is the targeted test for this; queued.

H3-extended robustness analysis (v1's H3 preserved)

the bio-vs-patient z-test on Bareche TIME / MC_global / MC_tumor labels (patient-level NMF subtypes from bulk RNA-seq) is preserved as the **H3-extended robustness analysis**:

run	TIME bio z	patient z	ratio	verdict (raw_he 0.129)
raw_he (floor)	4.79	37.25	0.129	baseline
raw_st (floor)	6.44	28.64	0.225	baseline
R6 (novae-only ST)	17.81	32.52	0.548	PASS
R1 (InfoNCE+late)	15.13	29.27	0.517	PASS
B2 (Procrustes)	12.66	39.83	0.318	PASS
R4 (InfoNCE+cross-attn)	10.07	22.97	0.438	PASS
R2 (SupCon+late)	8.14	27.01	0.301	PASS
R3 (Barlow+late)	6.18	25.03	0.247	PASS
B1 (CCA)	7.87	122.13	0.064	fail FAIL (anti-helpful)

every contrastive method passes; B1 is anti-helpful (amplifies patient 3.3x while only amplifying biology 1.6x). The original prediction “linear methods fail H3” was falsified by Procrustes passing – the failure mode is **correlation-maximization, not linearity**. CCA reweights to maximize between-modality correlation, inflating whichever shared axis is largest (patient identity in paired tumor data). Procrustes can only orthogonally rotate, so it preserves relative variance structure even as raw absolute deltas shift.

from 5 parents to 395 sub-pathways

the external review flagged that the 5-parent test was coarse-grained: broad parent unions blur mechanistically distinct programs. R4's specificity off-diag of 0.819 was consistent with that prediction. we extended the test from 5 top-level parents to **all 395 DAG nodes** under the 5 named parents (after gene-set size filter [3, 500]). for each node: gene set = union of descendant leaves' genes; AUCell scored per niche; univariate CCA per (run x view x DAG node), permutation null + BH-FDR over the 9480-cell grid.

option-A (cross-patient) significant cells per run

run	sig cells / 9480	%	% sig per parent (CC / ECM / Imm / PCD / TGF)
R4	934	9.9	69 / 93 / 83 / 75 / 81
B2	879	9.3	67 / 98 / 80 / 53 / 81
B1	832	8.8	58 / 98 / 76 / 60 / 71
B3	763	8.0	65 / 95 / 66 / 40 / 57
R6	437	4.6	38 / 82 / 33 / 32 / 33
R1	321	3.4	24 / 33 / 29 / 22 / 24
R2	273	2.9	low across all parents
R3	218	2.3	floor

R4 wins or ties on every parent. classical baselines (B1/B2/B3) catch up significantly at fine pathway resolution (95-98% on ECM, 53-80% on Immune) compared to the 5-parent test where the contrastive-vs-classical gap was decisive. R4 9.9% vs B2 9.3% vs R1 3.4% at DAG resolution is still a clear R4 lead but tighter than the 15-vs-5 spread at the 5-parent test. **R1/R6 near floor at DAG resolution** despite winning H2 niche-level biology preservation.

R4 top hits are biologically distinct sub-programs

pathway node	category	depth	R4 z_A
Co-inhibition by BTLA (R-HSA-9927353)	immune checkpoint	3	6.92
Interleukin receptor SHC signaling (R-HSA-912526)	cytokine signaling	4	6.80
Integrin cell surface interactions (R-HSA-216083)	ECM-cell adhesion	1	6.21
ECM proteoglycans (R-HSA-3000178)	ECM structural	1	5.73
Regulation of IFNA/IFNB signaling (R-HSA-912694)	interferon response	4	5.71
Degradation of extracellular matrix (R-HSA-1474228)	ECM remodeling	1	5.47
Elastic fibre formation (R-HSA-1566948)	ECM connective	1	5.33
Collagen biosynthesis (R-HSA-1650814)	ECM collagen	2	5.28
Collagen degradation (R-HSA-1442490)	ECM remodeling	2	5.22
FLT3 signaling via SRC family kinases (R-HSA-9706374)	tyrosine kinase	3	5.15
Calcineurin activates NFAT (R-HSA-2025928)	T-cell activation	4	5.10

these are biologically distinct sub-programs, not a single aliased axis. R4 separates immune checkpoint biology (BTLA, IL/SHC, IFN-alpha/beta, FLT3, calcineurin/NFAT) from ECM biology (integrin, proteoglycans, elastic fibre, collagen biosynthesis / degradation / trimerization). these map to clinically meaningful TNBC dimensions: immune checkpoint expression (relevant to checkpoint inhibitor response) and stromal/collagen remodeling (relevant to TME architecture and metastasis).

refined R4 aliasing framing

the 5-parent specificity off-diag 0.819 result holds at the 5-parent test but is a **coarse-aggregation artifact**, not a fundamental property of R4's representation. at DAG resolution R4 encodes biology at sub-pathway level. correct framing:

- at the **5-parent CCA test**: R4's directions appear aliased (off-diag 0.819) because parent unions of overlapping Reactome subtrees are themselves correlated. R4's tight manifold captures the shared variance underneath.
- at the **395-DAG-node CCA test**: R4 encodes distinct sub-programs (BTLA checkpoint vs collagen biology are clearly separable in its z_{he} projections).

artifacts: runs/tnbc-92/eval/H3/pathway_cca/dag_full/ - dag_metadata.parquet (395 nodes), per_node_aucell.parquet (286k niches x 395 cols), per_node_cca.parquet (9480 cells x option A + B + perm-null + BH-FDR).

the final read across H1, H2, H3

VERDICT **PASS under cross-patient generalization (the discriminating test).**
Proposal-stated test (Part A): all 8 runs **PASS**; every embedding encodes the 5 named Reactome pathway directions above null. **Discriminating test (Part B cross-patient):** R4 dominant (15/15); classical baselines strong (12-14/15); late-fusion contrastive fail (5/15). **Mechanism:** R4's tight ~3-dim manifold learns transferable representations of fine biology. at parent-subtree resolution its 5 canonical directions look correlated (off-diag 0.819) because the parent unions share gene content; at 395-DAG-node resolution R4 distinguishes specific sub-programs (BTLA, IFN, integrin, collagen biology) and still wins or ties on every parent. cross-attention's ST-conditional tile selection is the architectural ingredient; the apparent aliasing was a test-side artifact, not an encoding property. **Limitation:** at the 5-parent test resolution R4's canonical directions are correlated; tasks that want to read the 5 named parent pathways as 5 distinct axes should use R1/R3/B2 (more distinct at parent level) or query R4 at sub-pathway resolution where its distinctions are visible.

integrated H1 / H2 / H3 picture

metric	R1 / R6 (late-fusion contrastive)	R4 (cross-attention bridge)	B1 / B2 / B3 (classical)
H1 retrieval (AUC)	0.733-0.741	0.851	0.44-0.70
H1 structure (CKA)	0.23-0.25	0.564	0.09-0.12
H2 archetype ARI (patient-confounded)	mid	low (patient suppression)	win (patient leakage)
H2 MC ARI (niche-level KMeans)	win (0.23-0.25)	low (geometry bias)	mid (lost 31-37% on shift)
H2 LP MC accuracy (geometry-agnostic)	parity (0.26)	parity (0.23)	parity (0.21-0.24)
H2 Part B contrasts	2-3 / 6 (TLS only)	0 / 6 (annotation-purity)	0-2 / 6
H3 fit-on-all CCA	all pass	all pass	all pass (B2/B3 narrow win)
H3 cross-patient CCA	fail (5/15)	win (15/15)	strong (12-14/15)
H3 DAG decomposition (9480 cells, 395 nodes)	near floor (R1 3.4%, R6 4.6%)	9.9% sig, wins / ties every parent	strong (B1 8.8%, B2 9.3%, B3 8.0%)
H3-extended (Bareche)	PASS	PASS	B1 FAIL ; B2 PASS (linear)

note on baselines: B4 (random-init MLPs, no training, AUC 0.491) is reported on the H1 results table on page 9 but is not included in this integrated picture because it is an untrained sanity floor, distinct from B1/B2/B3 (trained linear methods). B4 confirms that R1's AUC 0.741 advantage over chance comes from contrastive training, not from the MLP architecture alone.

architectural recommendation (task-conditional)

cross-modal retrieval + cross-patient pathway generalization -> R4 (cross-attention bridge). **niche-resolution biology preservation + per-pathway interpretability** -> R1 or R6 (late-fusion contrastive). **avoid classical baselines** for either objective (B1 anti-helpful on H3-extended; B2/B3 saturate Part A by linear capacity but fail H1).

this is the omicstra alignment-agent's task-conditional routing rule: cross-patient/retrieval routes to R4; niche-resolution-interpretation routes to R1/R6.

proposal commitments, extensions, gaps, cross-confirmations

the proposal text (page 5) defines H1, H2, H3 with explicit metrics and baselines. this section maps what we delivered against that scope, what extends beyond it, what is missing and how much it matters, and where independent tests confirm the same conclusion.

table 1: proposal commitments and delivery status

hypotheses	proposal requirement	delivery	location
H1 base 1	CCA projection	done	runs/B1
H1 base 2	late fusion concat (non-contrastive)	done as B4 (random-init MLPs, R1 arch, no training; AUC 0.491, gap -0.001)	runs/B4/metrics_h1_raw.json
H1 base 3	raw unaligned concat	partial (B3 is PCA-512+L2, not 1856d raw)	runs/B3
H1 metrics	R@K, MRR, median rank, AUC, gap, CKA	done with rank-matched CKA control	eval/H1/summary.json
H1 range	predicted similarity 0.65-0.75	met for R1/R6/R2/B2; R4 above; R3 below	brief page 9
H2 A	ARI + silhouette on 9 archetypes vs single mod	done with caveat (patient-confounded)	eval/H2/summary.json
H2 B	per-compartment cosine, distinct > ambiguous	done, 1/6 contrasts pass (TLS only)	eval/H2/named_contrasts.parquet
H3	per-pathway CCA on 5 named Reactome	done with perm null + BH-FDR	eval/H3/pathway_cca/
H3 maps	spatial pathway-morphology maps	done (38 test subarrays, R4 z_he projection vs AUCell; spearman 0.34-0.56 per pathway)	eval/figures/pathway_spatial_maps/

extensions beyond proposal scope

table 2: extensions beyond proposal scope

extension	purpose	added when
4 contrastive variants (R1-R4)	isolate loss-function axis	original
R6 ablation (gpath2vec dropped from ST)	circularity check	original
rank-matched CKA control	falsifier for collapse-driven CKA inflation	step 3 review
H2 Part A-MC niche-level	proper biology test after label-resolution audit	label audit
H2 diagnostics (LP / kNN / rank-matched KMeans)	metric-independent triple-check on R4	geometry diagnostics
H3 cross-patient sub-split	discriminating test for proposal-spec H3	cross-patient test
H3 perm null + BH-FDR (1000 perms x 120 cells)	null calibration	cross-patient test
H3 specificity matrix (5x5)	mechanism diagnostic at 5-parent resolution	cross-patient test
H3-extended Bareche z-test	robustness check at patient-subtype level	step 3 review
B4 random-init MLP baseline	proposal-strict non-contrastive late-fusion (R1 arch, untrained)	v2.1
H2 Part B Welch z-test (6 named contrasts)	directional falsifier test on compartment cosine	H2 review
H3 spatial pathway-morphology maps	R4 z _{he} projection vs AUCCell, 38 test subarrays	v2.1
H3 DAG decomposition (395 sub-pathway nodes, 9480 cells)	extend H3 from 5 parents to full DAG, refines aliasing finding	DAG test

gaps with priority and cross-confirmations

table 3: nice-to-have and missing items

item	closes	cost	P
spatial pathway-morphology maps	H3 proposal deliverable, not currently generated	low	1
pure non-contrastive late-fusion concat baseline	H1 strict baseline 2 - does the loss help vs late-fusion arch alone?	med	1
stricter raw 1856-d unaligned concat baseline	H1 strict baseline 3	low	2
inter-pathway correlation matrix on AUCell-5	quantifies collinearity R4 aliasing exploits	very low	2
cell-composition partial-out on H3 specificity	tests if R4 aliased axis = cell composition or distinct biology	low	2
PROGENy signed-weighted pathway footprint	AUCell unsigned; adds direction + gene weights	med	2
18-class compartment vocabulary re-run (90-sub subset)	recover Part B contrasts inverted on 12-class collapsed vocab	low	2
R6b mirror ablation (gpath2vec-only ST)	symmetric R6 ablation; explains $z_{st} > z_{he}$ direction	med	2
AnInfoNCE (R5) per-dim temperature	addresses R4 dim-collapse aliasing trade-off	med	3
domain-adversarial GRL on patient_id	addresses R1/R6 cross-patient generalization failure	med-high	3

table 4: cross-confirmations (independent triangulation)

claim	independently confirmed by
R4's cross-modal advantage is real architectural work, not manifold compression	(1) H1 AUC 0.851+CKA 0.564 best in grid; (2) rank-matched CKA $R1@rank3=0.234$ vs $R4\ full=0.564$ gap -0.331; (3) $R4@rank3 = R4\ full$ (sanity)
H2 Part A archetype ARI is confounded by patient identity	(1) ARI ranking tracks patient-z exactly ($B1=0.307$ wins, $R4=0.059$ lowest); (2) 30/30 subarrays have archetype_unique=1; (3) classical baselines lose 31-37% on shift to mc_megacluster
R4 has niche-level biology despite KMeans failure	(1) linear probe $R4=0.230$ at parity with $R1=0.260$, raw_he=0.239; (2) rank-matched KMeans $R1@rank3=0.193$ vs $R4=0.098$ (half geometric); (3) H3-ext z_{he} TIME amp 2.1x
R4's 5-parent canonical directions are correlated, but R4 distinguishes sub-pathways at DAG depth	(1) at 5-parent: specificity off-diag mean =0.819 (max 0.970); (2) at 395-DAG-node: R4 leads (9.9% sig of 9480) and top hits are biologically distinct (BTLA, IFN, integrin, collagen); (3) parent unions share gene content - correlation is a test-side property, not an encoding property
Late-fusion (R1/R6) preserve niche biology but fail cross-patient	(1) H2 Part A-MC KMeans: $R6=0.251$, $R1=0.231$ winners; (2) H2 LP: $R1=0.260$ highest; (3) H3 cross-patient: $R1=5/15$, $R6=11/15$ (z_{st} only)
B1 CCA is anti-helpful for cross-modal interpretation	(1) H1 AUC 0.541 worst classical, CKA 0.086 worst overall; (2) H3-ext ratio $0.064 < raw_{he}$ floor 0.129; (3) loses cumulative correlation to rotation-equivalent B2/B3 despite optimizing for it

Procrustes B2 is a linear method that passes H3, falsifying "linear methods fail" framing	(1) H3-ext ratio 0.318 > raw_he 0.129; (2) H3 Part B 14/15; (3) reframes axis as correlation-maximization vs orthogonal rotation, not linear vs non-linear
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implications

(1) proposal commitments closed in v2.1: the random-init MLP baseline (B4) and the spatial pathway-morphology maps for R4. (2) the external review's predicted finding (parent-union collinearity) is confirmed both by the 5-parent specificity result (off-diag 0.819) and by the DAG decomposition showing that R4 distinguishes sub-pathways at fine resolution, so the parent-level correlation is a property of the test gene-sets, not a property of the encoding. (3) HESCAPE (Gindra 2025, ICCVW) provides field-level support for our R1/R6 cross-patient failure pattern, including at fine DAG resolution. (4) AUCell is unsigned and unweighted (Holland 2020; Wang+Thakar 2024). PROGENy with signed and weighted gene weights is queued as a future H3 layer for direction-aware pathway interpretation. (5) table 4 cross-confirmations is the strongest evidence layer, with every load-bearing claim independently supported by 2-3 tests.

not-yet-tried alignment methods + pre-registered predictions

three alignment-method extensions are queued. each addresses a specific finding from H1/H2/H3. predictions are pre-registered here so the same falsifier discipline that governed the H2 and H3 work carries forward to these runs.

AnInfoNCE (R5) – per-dim learnable temperature

Rusak et al., AISTATS 2025. Adds a learnable diagonal Λ into the InfoNCE similarity, allowing each dimension its own temperature. Targets the dim-utilization side of the R4 dim-collapse trade-off.

predicted result if run: AUC at least at R4's 0.851 with effective rank closer to R1 (~5-10 instead of R4's ~3). predicts effect on the 5-parent specificity off-diagonal (currently 0.819 for R4): if dim-collapse is what drives parent-level correlation, AnInfoNCE should bring R4's off-diagonal closer to R1's 0.481. Note that R4 already distinguishes sub-pathways at DAG depth, so the value of R5 is in parent-level specificity, not in DAG-level transfer (where R4 already wins).

caveat: per-dim learnable temperature targets dim utilization specifically. R4's parent-level correlation may also be partly a property of the test (gene-set overlap between Reactome parent unions) rather than the encoding, in which case AnInfoNCE will leave it largely unchanged. worth running; predicted effect size moderate.

R6b mirror ablation – gpath2vec-only ST

Symmetric to R6 (which dropped gpath2vec from ST, leaving novae 64d). R6b drops novae instead, leaving gpath2vec 512d only on ST. Directly tests whether R6's counter-intuitive $z_{st} > z_{he}$ pattern on H3 Part B is gpath2vec-driven or novae-driven.

predicted result: if z_{he} recovers under R6b (i.e. R6b z_{he} passes cross-patient where R6 z_{he} fails 4/5), the "InfoNCE without pathway anchor on ST forces z_{he} to mimic z_{st} " mechanism holds. If R6b z_{he} still fails, the mechanism is novae-side specific (Novae's GAT spatial structure is the actual implicit pathway anchor, not the explicit gpath2vec feature).

domain-adversarial GRL on patient_id

Adds a patient-classifier head with gradient reversal layer (GRL, Ganin & Lempitsky 2015) to actively suppress patient identity in the shared latent. Alternative path if cross-patient generalization fails on future cohorts (ex. HEST validation).

predicted result: would help R1/R6 close the H3 cross-patient gap (5/15 -> predicted 10-13/15) by penalizing the patient-specific gradient encoding directly. Trade-off: H1 AUC may drop because patient-correlated cross-modal signal is also suppressed.

HEST validation rationale: HESCAPE (Gindra 2025, ICCVW) reports field-level support for our R1/R6 cross-patient failure pattern across multiple ST cohorts. The GRL extension is the targeted intervention HESCAPE results endorse. queued for HEST validation when (package extraction) lands.

embedding spaces and what each metric measures

the H1/H2/H3 sections refer to z_{he} , z_{st} , z_{mean} , bio z , patient z , ARI, CKA, cosine, and manifold tightness throughout. these terms are overloaded in the multimodal-alignment literature. this section defines each precisely so the report's claims are self-contained.

the three embedding spaces

z_{he} , z_{st} , z_{mean} are not z -scores. they are the three output spaces of the alignment model. for every test niche i :

the three spaces, mathematically

z_{he_i} = output of the H&E-side MLP on input `virchow2_niche_i` (1280d). 512-d unit vector.

z_{st_i} = output of the ST-side MLP on input [`novae_niche_i` (64d) + `gpath2vec_niche_i` (512d)] = 576d -> 512d. 512-d unit vector.

z_{mean_i} = $(z_{he_i} + z_{st_i}) / 2$, then re-normalized to unit length. 512-d unit vector that averages information from both modalities.

L2 normalization. after each MLP, the output is divided by its Euclidean norm: $z_{hat} = z / ||z||_2$. so $||z_{hat}||_2 = 1$ for every embedding.

why L2 normalize. after normalization, the dot product $z_{hat_i} \cdot z_{hat_j}$ equals $\cos(\text{angle between } i \text{ and } j)$. cosine similarity becomes the natural distance metric. this is the standard contrastive-alignment protocol (CLIP, DINO, SimCLR, InfoNCE) - it removes magnitude effects and constrains embeddings onto the unit hypersphere where contrastive training is stable.

the three spaces can disagree. R6 (page 11) has spectacular z_{he} biology (TIME 17.81) but z_{st} biology collapses to 1.92. reporting only z_{mean} would hide this asymmetry. that is why all three views appear in the H3 results table.

bio z and patient z (the permutation z-scores)

the lowercase z in “z_he, z_st, z_mean” refers to the embedding space. the uppercase Z (or “z” in “bio z = 15.13”) refers to a permutation z-score computed on top of one of those spaces. these are different things using overlapping notation.

bio z and patient z, mathematically

pick one space (ex. z_he) and one label (ex. TIME).

step 1 (observed delta). partition niche pairs into same-label ($TIME_i = TIME_j$) and across-label ($TIME_i \neq TIME_j$). compute:

$$\begin{aligned} \text{delta_obs} = & \text{mean cos}(z_{\text{he}_i}, z_{\text{he}_j}) \text{ over same-label pairs} \\ & - \text{mean cos}(z_{\text{he}_i}, z_{\text{he}_j}) \text{ over across-label pairs} \end{aligned}$$

positive delta means same-TIME niches are closer in cosine than different-TIME niches.

step 2 (null distribution). shuffle the TIME labels 100,000 times. for each shuffle, recompute delta_k. this gives $\text{null} = \{\text{delta}_1, \dots, \text{delta}_{100000}\}$ under the hypothesis “label assignments are random.”

step 3 (z-score). $z = (\text{delta_obs} - \text{mean}(\text{null})) / \text{std}(\text{null})$. high z = the observed structure is far more extreme than chance.

patient z is the same calculation with patient_id as the label. the matched-null shuffle preserves patient cluster sizes during permutation so big patients don't artificially inflate the null.

so “R1 z_he TIME = 15.13” reads as: in run R1's H&E-side output space, niches sharing a TIME label are 15.13 standard deviations closer in cosine than would be expected under random label assignment. high bio z = strong biological organization. low patient z = good disentanglement from patient identity.

cosine similarity vs CKA: pairwise vs structural

these two metrics measure different things and a method can score well on one and poorly on the other.

cosine vs CKA, geometrically

cosine similarity = local pairwise geometry. are two specific embeddings pointing in similar directions? answers per-pair questions like “does H&E niche i match its paired ST niche i?” (this is what AUC and R@K measure).

CKA (Centered Kernel Alignment) = global relational geometry. compute the n-by-n Gram matrix K_{HE} of all pairwise cosines within the H&E space, and K_{ST} within the ST space. CKA measures whether these two relational structures look the same: does the overall shape of the H&E manifold match the overall shape of the ST manifold? answers structural questions like “if niches A and B are close in ST, are they also close in H&E?”

practical consequence. a system can have decent pairwise cosine retrieval (matched pairs cluster) but still fail to preserve broader manifold organization (relative neighborhoods don't match across modalities). that is why the report tracks both AUC/R@K and CKA - they are not redundant.

what “cross-modal structure” means

“the embedding space preserves cross-modal structure” means: relationships that exist in one modality are reflected similarly in the other after alignment. if niches A and B are close in ST space (both immune-rich), and B and C are close in ST (both stromal boundary), then their aligned H&E embeddings should preserve similar relative positions, not just match A_he to A_st pointwise. neighborhoods, topology, cluster organization, manifold shape - these should agree across modalities, not just paired points being close. that is why CKA matters beyond pairwise retrieval.

ARI, manifold tightness, and the two normalizations

ARI (Adjusted Rand Index)

ARI measures whether clustering in the embedding space matches a label structure. $ARI = 1$ means perfect agreement, $ARI = 0$ means random agreement. it is a single agreement scalar.

the H2 finding was that ARI on 9 per-patient archetype labels was **confounded by patient identity**. embeddings that clustered by patient (rather than by biology) scored high ARI because archetypes are themselves per-patient. that is why ARI was deprecated as the primary H2 metric in favor of per-compartment cosine on human-annotated morphology labels (which are tissue compartments, not patient identities, so cannot be confounded by patient leakage).

manifold tightness (R4 vs R1, geometrically)

“R4 forms a tighter manifold” means R4's embeddings collapse into fewer effective dimensions (~3 vs R1's ~5), with cross-modal matched pairs becoming geometrically compact. this is not collapse in the degenerate sense - structure concentrates into a dense shared latent regime with stronger global organization. R1 by contrast preserves broader cosine spread and potentially better retrieval diversity. so:

the R1 vs R4 regime trade-off

R1 = wider geometry. ~5 effective dimensions on z_{he} , ~10 on z_{st} . larger absolute cosine spread between bio-label groups. matched/mismatched cosines around 0.24/0.09 (gap 0.159). better z_{he} biology absolute. asymmetric across streams (z_{st} ratio only 0.155).

R4 = tighter shared manifold. ~3 effective dimensions on both z_{he} and z_{st} . matched/mismatched cosines around 0.60/0.41 (gap 0.195). best cross-modal alignment quality (AUC 0.851, CKA 0.564). best patient suppression. symmetric across streams (z_{st} ratio 0.397).

conclusion: regime trade-off, not winner-take-all. R1 is better for niche-level retrieval diversity, R4 is better for cohort-level group statistics. neither dominates as “more disentangled.”

L2 normalization vs z-scoring (different operations)

the report uses both, for different purposes. they are easy to confuse.

operation	formula	purpose	where used here
z-score	$x' = (x - \mu) / \sigma$ per feature, per subarray	remove subarray mean shifts; reduce batch/domain effects	Novae niche embeddings z-scored per subarray (helps Novae)
L2 normalize	$\hat{z} = z / \ z\ _2$ per vector	place embeddings on unit sphere; cosine becomes natural metric	output of every alignment MLP (R1-R4, B1-B3, R6) is L2-normed

encoder asymmetry finding. z-scoring helped Novae (cross-subarray silhouette improved -0.246 -> -0.135) but hurt Virchow2 (MC probe dropped 23.6% -> 10.4%). interpretation: Novae had unwanted subarray/domain shifts that needed removing; Virchow2's mean structure already encoded biologically meaningful morphology, so removing it destroyed signal. **encoder-specific preprocessing is the lesson.** L2 normalization is downstream of either choice and applies uniformly to all alignment outputs.

when modalities meet: early vs late integration

“fusion” refers to where in the pipeline the two modalities interact. this is a separate axis from loss function (InfoNCE vs Barlow vs SupCon) and is the architectural choice that distinguishes R4 (cross-attention) from R1/R2/R3/R6 (late MLPs).

the spectrum, schematically

strategy	where modalities interact	example
late fusion (two-stream)	each modality encoded independently; interact only at the loss after both have been projected to shared space	R1, R2, R3, R6 (independent MLPs; contrastive loss links them)
cross-attention bridge (modular early)	ST query attends over local H&E tile tokens BEFORE projection; modalities still mostly separate but interact at one interior layer	R4 (7 H&E tile tokens + ST query cross-attn, before MLP projection)
Novae-native early fusion (NOT done)	modalities jointly encoded from the start; tile tokens injected directly into Novae node features, joint message passing across modalities	not implemented; changes Novae architecture and interpretability

what we actually did: modular two-stream alignment

the core pipeline is fundamentally late fusion / two-stream alignment, not true early fusion. modalities are encoded independently first (virchow2_niche on the H&E side, novae+gpath2vec on the ST side), then aligned after projection into a shared 512-d space using a contrastive loss (InfoNCE / SupCon / Barlow). this is the classic late-interaction contrastive alignment pattern.

R4 introduces cross-attention as a **lightweight bridge** earlier in the pipeline, but the modalities still maintain separate representational pipelines before interaction. wording from the R4 design note: “*modular two-stream alignment, not Novae-native early fusion.*” R4 is structurally somewhere between R1 and a true early-fusion architecture.

R4 cross-attention bridge: what changed

cross-attention bridge, mathematically

R1 (late fusion). H&E side input is virchow2_niche, the mean-pooled Virchow2 embedding over center spot + 6 spatial neighbors (1280d). pooling happens BEFORE alignment.

R4 (cross-attention bridge). H&E side input is 7 raw virchow2_cell tile tokens (center + 6 neighbors), each 1280d. ST projects to a query vector q_{st} , then computes attention weights over the 7 tile tokens: $\alpha_k = \text{softmax}(q_{st} \cdot \text{tile}_k)$. H&E representation = $\sum_k \alpha_k \cdot \text{tile}_k$. ST query SELECTS which tiles matter for this niche, instead of mean-averaging them all.

parameter cost. ~4M extra params (small attention head + query projection). local token set (7 tiles), so quadratic cost is negligible. small architectural change, big structural effect.

why this matters biologically. mean-pooling can destroy local morphology information before alignment - necrotic edges, lymphocyte boundaries, glandular interfaces, stromal transitions, TLS morphology can disappear under mean pooling. cross-attention preserves local token heterogeneity and lets ST conditionally select which tile is morphologically relevant.

pros, cons, and the empirical evidence

strategy	pros	cons	empirical result
late fusion (R1, R2, R3, R6)	simple, modular; encoders pluggable; cheap compute; easy to ablate	pooling discards local morphology before alignment; wider manifold = lower cross-modal CKA	R1 AUC 0.741, CKA 0.242 passes H1 but lower than R4
cross-attention bridge (R4)	preserves local tile heterogeneity; conditional tile selection; tighter manifold; +0.11 AUC, +0.32 CKA	only ~3 effective dims (may hurt within-space retrieval diversity); ~4M extra params	R4 AUC 0.851, CKA 0.564 best H1 winner; only run moving toward bio-patient symmetric on both views
classical linear (B1, B2, B3)	closed-form, no training; fast; deterministic	no nonlinear capacity; CCA inflates patient axis; B3 negative gap (sanity)	B1 fails H3; B2 passes H3 (linear surprise); all weaker than contrastive on H1

future direction: where to inject IHC

extending the system with IHC biomarker embeddings (Ki67, CD8, PD-L1, FOXP3, PanCK) is biologically natural since IHC sits between morphology and transcription as a phenotype anchor. four plausible insertion points, ordered by engineering cost:

option	where	cost	scientific value
1. concat into ST stream (near-term default)	Novae 64d + gpath2vec 512d + IHC ~32-128d -> MLP_st	low (no arch change)	modular, easy to ablate; fits H1/H2/H3 framework
2. auxiliary supervision (strongest scientifically)	$L = L_{\text{InfoNCE}} + \lambda \cdot L_{\text{IHC}}$ (predict biomarker profile from shared latent)	medium (loss + head)	constrains latent space by phenotype, not just patient identity; reduces patient leakage
3. cross-attn bridge (future R5+)	H&E tiles <-> ST token <-> IHC biomarker token (three-way attention)	medium-high (arch extension)	IHC as interpretable bridge modality; morphology-grounded phenotype attention
4. true early fusion (invasive, not recommended)	inject IHC channels directly into Novae node features	high (changes Novae)	hard to interpret; hard to isolate effects; loses ablation cleanliness

recommended sequencing

near-term: option 1 (concat IHC into ST stream). preserves the current H1/H2/H3 evaluation framework, adds another ablation row (does z_{he} biology depend on IHC being in ST?), tests a clean question.

mid-term: option 2 (IHC as auxiliary supervision). this is the most scientifically interesting direction. constraining the latent space by transcriptomics + morphology + protein phenotype simultaneously could reduce patient leakage further while preserving biological structure - directly answering the disentanglement question the report is built around. **do not** replace Novae with IHC: Novae captures spatial graph topology and transcriptomic manifold organization that IHC cannot; IHC is a conditioning signal and phenotype anchor, not a primary graph encoder.

pre-registration framework

the H2 and H3 work was governed by an explicit pre-registration discipline. this section documents the protocol so it can be applied uniformly to future cohorts (HEST validation) and so reviewers can audit the falsifier chain.

five operating principles

1. predictions locked in memory before computing. each test stage pre-registered specific scenario predictions in dedicated memory files (`project_h2_mc_coherence_predictions.md`, `project_h3_pathway_cca_predictions.md`, `project_h2_mc_diagnostics_predictions.md`) before any compute. predictions specified the expected ranges per run, the falsifier criteria, and the interpretation rule for each possible outcome. this prevents post-hoc rationalization of surprising results.

2. falsifier outcomes documented honestly. The initial niche-level H2 test result surprised the project by ranking R4 lowest on K-means MC ARI. user instinct that “the metric is wrong” triggered the metric-independent metric-independent diagnostics. the linear probe (the load-bearing diagnostic) showed R4 at parity with R1/R6. the K-means MC ARI result was preserved in the report as evidence the metric was misleading, NOT archived. preserving surprising results alongside their resolutions is what makes the methodology defensible.

3. falsifier triggers fired and were resolved. the H3 cross-patient test pre-registered 4 falsifier triggers. **3 of 4 fired:** B2/B3 tied B1 on cumulative correlation (CCA losing on its own optimization target); R4's 5-parent specificity matrix had high off-diagonal (0.819) rather than distinct directions; option A vs B agreement showed sharp divergence (B1 $r=0.146$, R4 $r=0.630$). the initial reading from these surprises was that R4 wins cross-patient because it aliases. the DAG decomposition (395 sub-pathway nodes, 9480 cells) ran later as the next falsifier step and overturned the aliasing-as-mechanism reading: R4 distinguishes BTLA, IFN, integrin, and collagen sub-programs at fine resolution and still wins or ties at every parent. the 5-parent off-diagonal was a property of the gene-set unions, not the encoding.

4. R4's surprising win does not get a special interpretation. the same falsifier framework that demoted B1 on H3-extended (anti-helpful ratio below raw floor) is what promoted R4 on H3 cross-patient. The 15/15 cell win at $z=3.8-5.7$ is exactly the kind of result the pre-registration was designed to make robust. R4 didn't get a pass because it's the “preferred” method; it got promoted because the discriminating test ranked it dominant. when the DAG analysis falsified the initial aliasing-as-mechanism story, the report was revised rather than the result. that is the falsifier framework working correctly.

5. methodology arc preserved as a project artifact. `project_h2_2_5_2_6_methodology_arc.md` documents the general lesson: when a single-metric result violates strong priors from multiple other metrics, stop and run metric-independent diagnostics before integrating the result into downstream verdicts. this is a transferable lesson, not a TNBC-92-specific finding. it will apply to HEST validation, future cohorts, and any future contrastive-alignment evaluation in the omicstra package.

the protocol, summarized

1. pre-register predictions per scenario before computing. 2. run primary metric. 3. if surprised (result violates multiple priors), do NOT integrate; run metric-independent diagnostics. 4. integrate verdict only after diagnostics converge or formally diverge. 5. preserve both the surprising result and the diagnostic resolution in git/memory.

three hypotheses, three verdicts

H1 . alignment quality

do learned methods couple H&E and ST?

PASS

all 4 contrastive runs beat all 3 classical baselines on AUC and CKA-after. R4 (InfoNCE + cross-attention) leads by a wide margin, and the rank-matched CKA control rejects the manifold-collapse hypothesis ($R1@rank3 = 0.234$ vs $R4$ full = 0.564, gap -0.331).

H2 . structure preservation

does the shared space preserve compartments?

PARTIAL -> fusion-axis trade-off

the original H2 metric (ARI on 9 archetypes) turned out to be a patient classification test in disguise: archetype labels are assigned per patient, so KMeans on aligned embeddings recovers patient identity, not within-patient biology. the niche-level proper test uses `mc_megaccluster` (14 classes, per-spot NMF). result: R1/R6 win KMeans MC ARI (0.231-0.251); R4 scores low on KMeans (0.098) but **passes the patient-stratified linear probe at parity** with R1/R6/raw_he. R4's KMeans score reflects KMeans's bias against tight low-dimensional manifolds, not loss of biology. Part B compartment cosine: 1 of 6 contrasts supported (TLS vs TIL only); the other 5 inverted because the Tumor and Necrosis annotations pool subtypes (Tumor $n=4384$, TIL $n=105$) and dilute matched-pair cosine. **fusion-axis verdict**: late-fusion contrastive (R1, R6) for niche-level biology with separable cluster regions; cross-attention bridge (R4) for cross-modal alignment with tighter manifold geometry. neither family dominates on every metric.

H3 . pathway interpretability

do 5 named Reactome pathways correlate with shared latent directions?

PASS under cross-patient generalization

proposal-stated test (Part A): every run passes BH-FDR; the proposal hypothesis is supported universally but the test is saturated by 512-d projection onto 1-d targets. **discriminating test (Part B cross-patient)**: R4 dominant 15/15. classical baselines strong 12-14/15. late-fusion contrastive (R1/R2/R3) fail at 5/15. **DAG decomposition (395 sub-pathway nodes)**: R4 still leads (9.9% of 9480 cells) and wins or ties on every parent. top R4 hits are biologically distinct (BTLA checkpoint, IFN-alpha/beta, integrin adhesion, collagen biosynthesis and degradation, NFAT signaling), so the 5-parent off-diagonal of 0.819 reflects gene-set overlap between Reactome parent unions, not a property of R4's encoding. classical baselines stay competitive at fine resolution (B1/B2 reach 98% on ECM); R1/R6 fall to floor (3-5%) at DAG depth. **H3-extended robustness (Bareche labels)**: every contrastive method passes; only B1 fails (ratio 0.064 below raw floor). Procrustes (B2) passes despite being linear: the failure mode is correlation-maximization, not linearity.

architectural recommendation (task-conditional): R4 for cross-modal retrieval and cross-patient pathway transfer at any DAG depth. R1/R6 for niche-level clustering (H2). classical baselines (B1/B2) acceptable for fine ECM and Immune sub-pathway detection; avoid B1 for any biology-over-patient interpretation task.